Toulouse School of Economics — M2 Econometrics & Statistics

Geomarketing Retail Location Analysis

Predicting Optimal Locations for New Retail Stores

Using gravity models, principal trading areas and spatial econometrics

Course	Geomarketing — 2025-2026
Institution	Toulouse School of Economics (TSE)
Programme	M2 Econometrics & Statistics
Project	Final Project — Retail Location Analysis
Team	Alexandre Picart Badiallo Kantako
	Dorian Senseby Lucien Mirlicourtois
Date	April 2026
Group	Mbappe67

Code: project shared directory | Shiny app: interactive Leaflet of best & worst market zones

Contents

1 Context and Strategic Objectives **3**

2 Data preparation and Feature Creation **3**

3 Delimitation of Principal Trading Areas (PTA) **4**

4 Econometric Modeling: Specifications and Selection **4**
4.1 Selected Model for Sales Volume (Model V2) 5
4.2 Selected Model for Market Share (Model P1) 5
4.3 Model Comparison & Selection 6

5 Strategic Recommendations and Candidate Evaluation **6**

6 Evaluation of 10 Candidate Locations **6**
6.1 Selection Methodology 6
6.2 Results — Predicted Performance 7
6.3 Best & Worst Candidate — Criteria 8

7 Conclusion **8**

References **9**

Detailed Report: Retail Location & Predictive Analysis

1. Context and Strategic Objectives

In today's highly competitive retail landscape, the physical location of a store remains a paramount driver of commercial success. The objective of this project is to analyze the sales data of a retail client in order to (i) characterize the principal trading areas of existing stores, and (ii) identify the most promising locations for future store openings.

Choosing where to open a new store involves significant financial commitment. A poor location can lead to underperforming assets or, worse, cannibalize sales from existing stores. Conversely, an optimal location captures unexploited demand and maximizes Return on Investment. This analysis serves to mitigate these real estate decisions. By auditing the performance of the current network and quantifying the spatial and socio-economic determinants of consumer behavior, we provide actionable intelligence. This predictive approach empowers the company prioritize the most profitable geographical areas for future store openings.

To achieve this, the study fuses heterogeneous databases: proprietary client data (sales history, store coordinates), the SIREN registry to map competitive pressure, socio-demographic data at the IRIS (statistical neighborhood) level provided by the French statistics bureau (INSEE), and OpenStreetMap (OSM) data to quantify local attractiveness via Points of Interest (POIs).

2. Data preparation and Feature Creation

The accuracy of a gravity-based model relies heavily on the granularity and quality of the engineered data. Our analytical framework operates across three distinct levels:

- **The Store Level:** For each existing store, we aggregated pure performance indicators (total revenue, customer footfall). Spatial context variables were then engineered. Competitive pressure was measured by counting direct competitors (via SIREN) within a 2-kilometer radius. Concurrently, site attractiveness was estimated by counting POIs (parking lots, transit stops, adjacent retail) within a 5-kilometer radius, serving as a proxy for natural pedestrian and vehicular traffic.

- **The Demand Zone Level (IRIS):** Every customer was geolocated and linked to their IRIS of residence. This facilitated the calculation of market penetration rates per neighborhood, defined as the ratio of captured revenue to the total estimated market potential of that zone. To refine behavioral analysis, socio-demographic proxies were integrated: the proportion of managerial/executive households (a proxy for purchasing power and brand positioning), the proportion of young residents, and the share of detached houses (an indicator of urban morphology and mobility).

- **The Interaction Level (IRIS $\times$ Store Pair):** This is the fundamental unit of observation. Each row represents the commercial flow between a residential zone and a specific store. The key variable is the Euclidean distance (calculated using the Lambert 93 projection to ensure metric accuracy). Market share was also computed at this level, representing the fraction of an IRIS's total expenditure captured by a specific store.

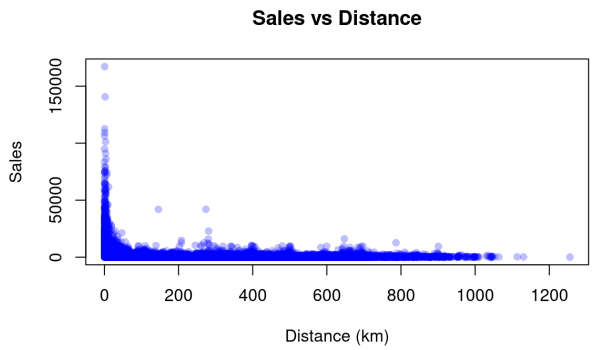

Figure 1: Scatter plot Sales - Distance

3. Delimitation of Principal Trading Areas (PTA)

A crucial step in retail analysis is defining a store's true geographical sphere of influence. Following the canonical standards of academic literature (Applebaum, 1966; Huff, 1964), we defined the Principal Trading Area (PTA) as the geographic perimeter generating **80% of the store's total sales**.

This 80% threshold is not arbitrary. It effectively isolates the core customer base while systematically excluding opportunistic consumers (e.g., tourists or long-distance commuters) who would otherwise artificially inflate the zone's size and distort local analysis. The construction algorithm is systematic: for each store, contributing IRIS zones are sorted in descending order of revenue generation. A cumulative sum is calculated until the 80% threshold is breached, after which the geometries of these specific IRIS are merged into a continuous, mappable polygon.

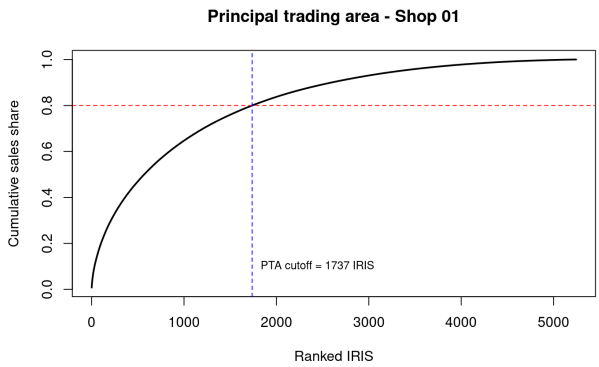

Figure 2: Example for Principal Trading Areas of Shop 01

4. Econometric Modeling: Specifications and Selection

To ensure the robustness of our predictions, three specifications were tested for sales volumes (V1: Quasi-Poisson, V2: Log-Linear OLS, V3: OLS with interaction) and three for market shares (P1: OLS Logit, P2: Beta Regression, P3: Weighted Binomial Logit).

In both instances, the final model selection was based on the specification that maximized explanatory power measured by the **Adjusted** R^2, while preserving clear business interpretability. Standard errors were clustered at the IRIS level to account for spatial autocorrelation.

4.1 Selected Model for Sales Volume (Model V2)

Model	Specification	Rationale
V1	Quasi-Poisson (GLM)	Handles overdispersion and count-like data. No closed-form R^2.
V2	Log-linear OLS	**Main prediction model.** Clustered SE by IRIS. Selected for predictions.
V3	Interaction ln(dist) × ln(POI)	Tests whether POI attractiveness attenuates distance friction.

Table 1: Sales volume models

Model V2 (Log-linear OLS) was selected as it provided a good fit to the data ($R^2 = 0.223$), outperforming the Quasi-Poisson and very closed to interaction models. The logarithmic transformation allows the coefficients to be directly interpreted as elasticities. The equation estimated at the IRIS i / Store j pair level is:

$$\ln(\texttt{Sales}_{ij}) = \beta_0 + \beta_1 \ln(\texttt{Dist}_{ij}) + \beta_2 \ln(\texttt{POI}_j) + \beta_3 \ln(\texttt{Comp}_j) + \sum \gamma_k X_{i,k} + \varepsilon_{ij}$$

Where $X_{i,k}$ represents the vector of socio-economic characteristics of IRIS i (population, proportion of executives, etc.).

The results reveal a distance elasticity of -0.378: a 10% increase in distance results in a 3.8% drop in predicted sales. Site attractiveness boosts sales (POI, $+0.135$), while competitive density naturally erodes them. Interestingly, the proportion of executive households negatively impacts absolute sales volume, highlighting a potential misalignment in brand positioning for upscale demographics.

4.2 Selected Model for Market Share (Model P1)

Model	Specification	Rationale
P1	Logit OLS	**Main share model.** OLS on logit(share). SE clustered by IRIS.
P2	Beta regression	MLE on share $\in (0, 1)$. More appropriate for bounded proportions.
P3	Binomial logit (weighted)	GLM weighted by number of customers. Accounts for IRIS size heterogeneity.

Table 2: Market share models

To model the proportion of local expenditure captured by a store (bound between 0 and 1), we utilized Huff's discrete choice framework (1964). Model P1 (OLS on a Logit transformation) outperformed the Beta regression, yielding the highest R^2 (0.263). The formula is:

$$\ln\left(\frac{\texttt{Share}_{ij}}{1 - \texttt{Share}_{ij}}\right) = \alpha_0 + \alpha_1 \ln(\texttt{Dist}_{ij}) + \alpha_2 \ln(\texttt{POI}_j) + \alpha_3 \ln(\texttt{Comp}_j) + \sum \delta_k X_{i,k} + \mu_{ij}$$

Distance remains the strongest deterrent to market share capture (-0.449). Furthermore, this model reveals that highly residential neighborhoods (high proportion of detached houses) and executive households—despite spending less in absolute volume—exhibit extreme loyalty and generate very high local market shares when they do choose to visit the store.

4.3 Model Comparison & Selection

Model	Target	Adj. R^2 / pseudo-R^2	Note
V2: Log-linear OLS	Sales	0.2229	**Selected for predictions**
V3: Interaction	Sales	0.2261	Marginal improvement over V2
P1: Logit OLS	Share	0.2629	**Selected for predictions**
P2: Beta regression	Share	0.2601	Consistent with P1
V1: Quasi-Poisson	Sales	N/A	Deviance-based; robust to zeros
P3: Binomial logit	Share	N/A	AIC = 228,588; consistent signs

Table 3: Model fit comparison

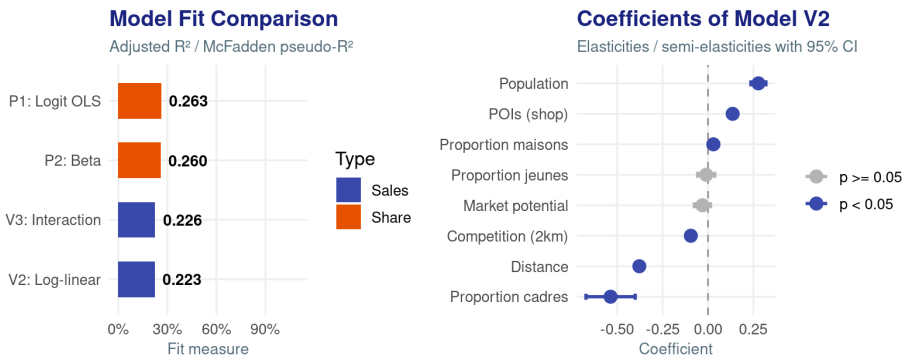

Figure 3: Model fit comparison

V2 is selected for sales predictions because it is the most interpretable, and it allows straightforward out-of-sample prediction for new locations. P1 is selected for market share for the same reasons. Both are estimated with IRIS-clustered standard errors to account for spatial dependence.

5. Strategic Recommendations and Candidate Evaluation

6. Evaluation of 10 Candidate Locations

6.1 Selection Methodology

To identify optimal future locations, we generated a list of potential sites drawn from the SIREN competitor database: a zone where competitors are present confirms that demand for this product category exists. We then focus on zones with the largest *uncaptured* potential.

We then filtered and ranked these candidates using a rigorous four-step methodology.

- **Criterion 1 - Existing Demand:** Strictly retain IRIS zones with a positive market potential ($mp > 0$).
- **Criterion 2 - Proven Market:** Require the presence of at least one direct competitor in the IRIS to confirm local consumer appetite.

- **Criterion 3 - Anti-Cannibalization Rule:** Exclude any candidate zone located within a 2-kilometer radius of an existing client store.
- **Criterion 4 - Final Ranking:** Surviving candidates are sorted in descending order based on their *uncaptured market potential* (Total Potential − Sales already captured by the company in that zone).

For each candidate, predictions use V2 (sales) and P1 (market share), setting shop characteristics to average existing values (`avg_poi`, `avg_compet_2km`). The 80% PTA is reconstructed with the same cumulative method as for existing stores.

6.2 Results — Predicted Performance

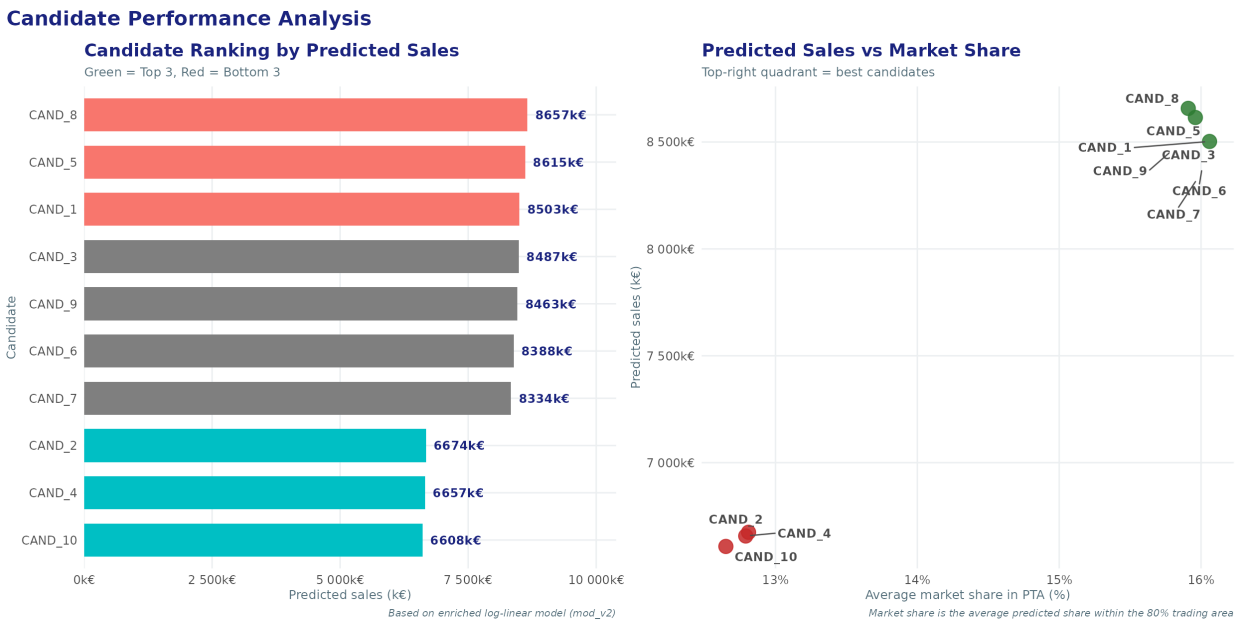

Figure 4: Candidate Performance Analysis on predicted sales

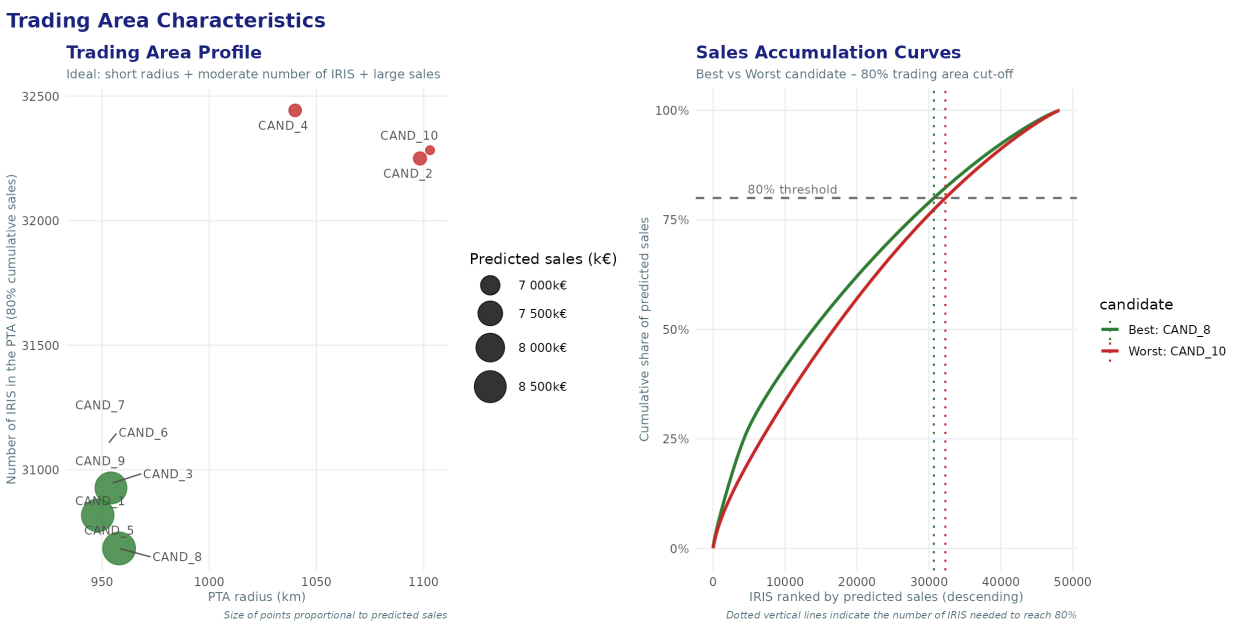

Figure 5: Candidate Performance Analysis on trading area

There is a clear break between ranks 1–7 (predicted sales ≈ 8.3–8.7M€, market share ≈ 15.8–16.1%) and ranks 8–10 (≈ 6.6M€, share ≈ 12.6–12.8%), reflecting a structural difference in market potential between the two groups.

6.3 Best & Worst Candidate — Criteria

CAND_8 — Best candidate (Rank #1) (In IRIS 916650000)

Predicted sales: **8,657,305 €** Market share: **15.91%** PTA radius: **958 km** IRIS in PTA: **30,684**

This site emerges as the absolute investment priority. The model predicts an annual sales volume of approximately **€8.66 million**. Its key advantages include:

- **Maximum uncaptured economic potential**: Almost the entirety of the zone's market potential is currently unserved by the company (over €4.3 million in uncaptured spend).
- **High predicted market share** (15.91%) relative to the zone's potential.
- **Highly compact trading area** (radius under 1 kilometer). A tight catchment area guarantees a hyper-local customer base that experiences minimal distance friction, ensuring high visit frequency and strong loyalty.

CAND_10 — Worst candidate (Rank #10) (In IRIS 441950000)

Predicted sales: **6,608,095 €** Market share: **12.65%** PTA radius: **1,103 km** IRIS in PTA: **32,283**

This location represents a high-risk investment. Predicted sales plateau at **€6.6 million**, roughly 25% lower than the top-tier candidates.

- **Lowest market share** (12.65%): the store attracts a smaller fraction of local spending.
- **Extremely dispersed trading area**: Reaching the 80% revenue threshold requires drawing customers from over 32,000 IRIS zones across a vast radius. The store would rely heavily on distant, volatile customers who are highly susceptible to competitor churn.

Three criteria drive the ranking: **(1) uncaptured market potential** — the primary economic rationale; **(2) predicted market share** — reflects competitive environment and site accessibility; **(3) PTA compactness** — a dispersed zone implies high distance friction and fragile customer loyalty. The best candidate combines all three: high potential, good share, compact zone.

7. Conclusion

This study allowed us to identify the best strategic location to open a new store and maximize the client's sales. To achieve this, we combined several databases: the purchase history and locations of current customers, the locations of competitors, as well as socio-economic information at the neighborhood level (IRIS).

Our methodology was highly rigorous. To be as accurate as possible, we tested and compared several predictive models widely used in geomarketing (gravity models). By selecting the ones that best explained the data (the highest R^2), we were able to clearly measure the major impact that distance and a neighborhood's attractiveness have on consumer choices.

The selection of our best candidate was not left to chance. It relies on strict filters: we targeted areas with proven demand (where competitors are already present), while avoiding placing a store too close to an existing one (to prevent cannibalization). Above all, we looked for areas with the highest uncaptured market share.

Thanks to this method, **candidate site CAND_8 (located in IRIS 916650000)** clearly stands out as the best choice for the next opening. It combines excellent potential revenue (€8.66 million), a dominant local market share, and a very compact geographic area that will make it easier to build customer loyalty.

Limitations of the analysis: It is important to note that the distances calculated in our models are straight lines ("as the crow flies"). For future analyses, using actual driving times on the road network would allow us to account for physical obstacles (rivers, highways) and further refine our predictions. Finally, adding more socio-economic variables would help us capture customer purchasing behaviors even more accurately based on their specific profiles.

Full R code: /data/cg2026/mbappe67 | Shiny app: https://badiallo.shinyapps.io/mbappe67/
Groupe Mbappe67— TSE M2 Econometrics & Statistics

References

- Applebaum, W. (1966). Methods for Determining Store Trade Areas, Market Penetration, and Potential Sales. *Journal of Marketing Research*, 3(2), 127–141.
- Huff, D. L. (1964). Defining and Estimating a Trading Area. *Journal of Marketing*, 28(3), 34–38.
- Reilly, W. J. (1931). *The Law of Retail Gravitation.* New York: Knickerbocker Press.